24/Aug/20

1. $f(t) = \sin t + \cos(t - \pi/4)\, u(t - \pi/4)$

$L\{F(t)\, u(t-a)\} = e^{-as} L\{f(t+a)\}$.

$L\{f(t)\} = \frac{1}{s^2+1} + e^{-\pi/4 s} L\{\cos(t)\}$

$\frac{1}{s^2+1} + e^{-\pi/4 s} \frac{s}{s^2+1}$

$\cancel{\frac{1}{s^2+1} + e^{-\pi/4 s} L\left\{\frac{\cos t}{\sqrt{2}} - \frac{\sin t}{\sqrt{2}}\right\}}$

$\cancel{\frac{1}{s^2+1} + \frac{e^{-\pi/4 s}}{\sqrt{2}} L\{\cos t - \sin t\}}$

$\cancel{\frac{1}{s^2+1} + \frac{e^{-\pi/4 s}}{\sqrt{2}} \left[\frac{1}{s^2+1} - \frac{s}{s^2+1}\right]}$.

Susanth

2. $t - t\,(u(t-1)) + u(t-1)$

$t - u(t-1)[t-1]$ $\quad L\{f(t-a)\, u(t-a)\} = e^{-as} F(s)$.

$L\{f(t)\} = \frac{1}{s^2} - \frac{e^{-as}}{s}$

Susanth

3. $f(t) = t + 3\,[u(t-2)]$

$(t-2)\, u(t-2) + 5\, u(t-2)$.

$L\{f(t-a)\, u(t-a)\} = e^{-as} F(s)$

$\frac{e^{-2s}}{s^2} + \frac{5e^{-2s}}{s}$

Susanth

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4. $1 - u(t-2) + t^2 \; u(t-2)$

$L\{f(t)\} = L\{1\} - L\{u(t-2)\} + L\{t^2 \; u(t-2)\}$.

$= \frac{1}{s} - \frac{e^{-2s}}{s}$

$L\{f(t) \; u(t-a)\} = e^{-as} L\{f(t+a)\}$

$= e^{-2s} L\{(t+2)^2\}$

$= e^{-2s} L\{t^2 + 4 + 4t\}$

$= e^{-s} \left[\frac{2}{s^3} + \frac{4}{s} + \frac{4}{s^2} \right]$ Sourabh

5. $u(t-1)$

$\frac{e^{-s}}{s}$. Sourabh.

6. $u(t-\pi) - u(t-2\pi)$

$\frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}$ Sourabh.

7. $e^t \, u(t-4) - e^t \, u(t-5)$.

$L\{f(t) \; u(t-a)\} = e^{-as} L\{f(t+a)\}$

$e^{-4s} L\{e^{t+4}\} - e^{-5s} L\{e^{t+5}\}$

$e^{-4s} \frac{e^4}{s-1} - e^{-5s} \frac{e^5}{s-1}$ Sourabh

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8. $\cancel{\frac{e^{-3s}}{s+3}} - \cancel{\frac{e^{-4s}}{s-4}}$ $\quad \frac{e^{-3s}e^{4}}{s-1} - \frac{e^{-4s}e^{4}}{s-1}$

Swarna

9. $u(t-5)\,\frac{t-5}{5} - u(t-10)\,\frac{t-5}{5} + u(t-10)$ Swarna

$\frac{1}{5}[u(t-5)\,t] - u(t-5) - \frac{u(t-10)\,t}{5} + u(t-10) + \cancel{u(t-10)}$

$L\{f(t)\,u(t-a)\} = e^{-as}\,L\{f(t+a)\}$.

$\frac{1}{5}e^{-5s}L\{t+5\} - \frac{e^{-5s}}{5} - \frac{1}{5}e^{-10s}L\{t+10\}$

$\frac{1}{5}e^{-5s}\left(\frac{1}{s^2}+\frac{5}{s}\right) - \frac{e^{-5s}}{5} - \frac{1}{5}e^{-10s}\left[\frac{1}{s^2}+\frac{10}{s}\right]$.

Swarna

10. $t - t\,u(t-2) + t^2u(t-2) - t^2u(t-6) + t^3u(t-6)$

$\frac{1}{s^2} - L\{t\,u(t-2)\} + L\{t^2\,u(t-2)\} - L\{t^2\,u(t-6)\} + L\{t^3\,u(t-6)\}$

$L\{t\,u(t-2)\} = e^{-2s}L\{t+2\}$
$= e^{-2s}\left(\frac{1}{s^2}+\frac{2}{s}\right)$.

$L\{t^2u(t-2)\} = e^{-2s}L\{(t+2)^2\}$
$e^{-2s}L\{t^2+4t+4\}$
$e^{-2s}\left[\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right]$

$L\{t^2u(t-6)\} = e^{-6s}L\{t^2+36+12t\}$
$= e^{-6s}\left[\frac{2}{t^3}+\frac{12}{s^2}+\frac{36}{s}\right]$

$\begin{array}{r} 36 \\ \times 6 \\ \hline 216 \end{array}$

$L\{t^3u(t-6)\} = e^{-6s}L\{(t+6)^3\}$
$= e^{-6s}L\{t^3+216+3t\times36+3t^2\times6\}$
$= e^{-6s}L\{t^3+216+108t+18t^2\}$
$= e^{-6s}\left[\frac{6}{s^4}+\frac{216}{s}+\frac{108}{s^2}+\frac{18\times2}{s^3}\right]$

$\begin{array}{r} 36 \\ \times 3 \\ \hline 108 \end{array}$

Swarna